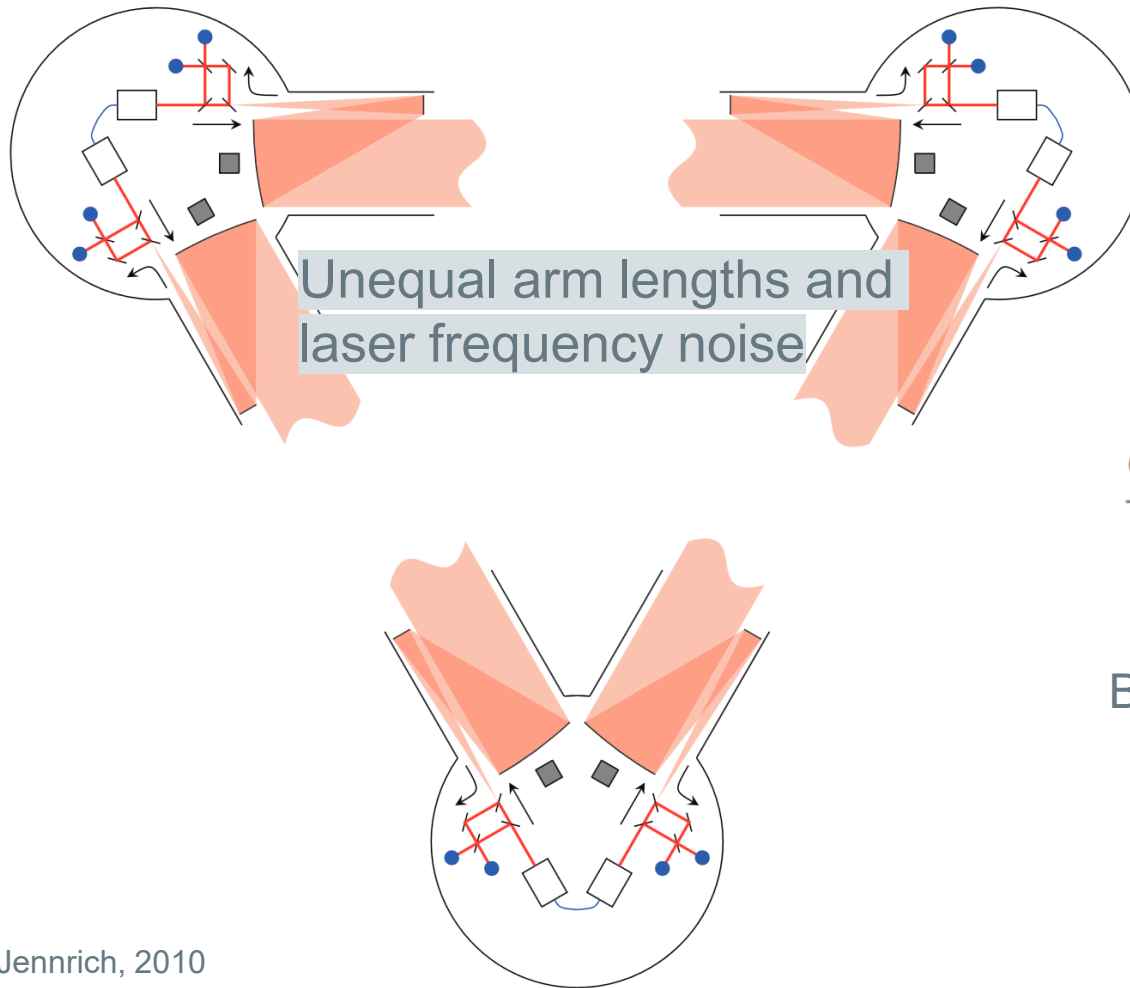




Testbed for Validating Second-Generation TDI and Clock Noise Correction for LISA

Karin Kruuse, Reid Ferguson, Patrick Krieger, Guido Mueller

The LISA Satellites



As per LISA requirements

$$\frac{\delta\nu}{\nu} = 10^{-13}$$

Hence for the arm length uncertainty we get

$$\frac{\delta\nu}{\nu} = \frac{\delta L}{\Delta L} \Rightarrow \delta L \approx 10^{-13} \cdot 10^7 \text{ m} = 1 \mu\text{m}$$

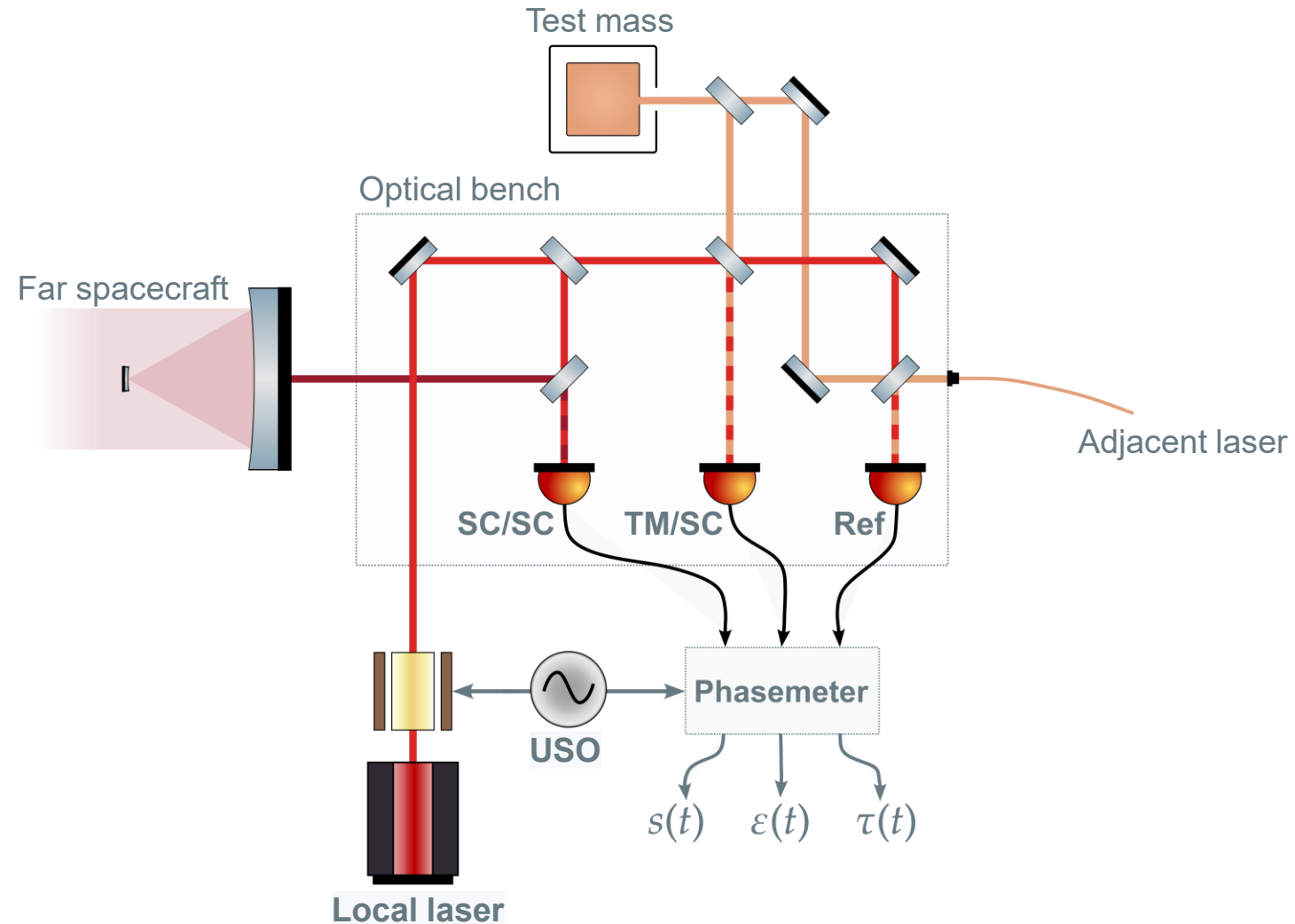
But to be sensitive enough to detect GWs, we need

$$\delta L = 10^{-13} \text{ m} \Rightarrow \Delta L = 1 \text{ m}$$

LISA Interferometric Metrology System



MAX-PLANCK-INSTITUT
FÜR GRAVITATIONSPHYSIK
(Albert-Einstein-Institut)



USO – Ultra Stable Oscillator

$s(t)$ – science beat note

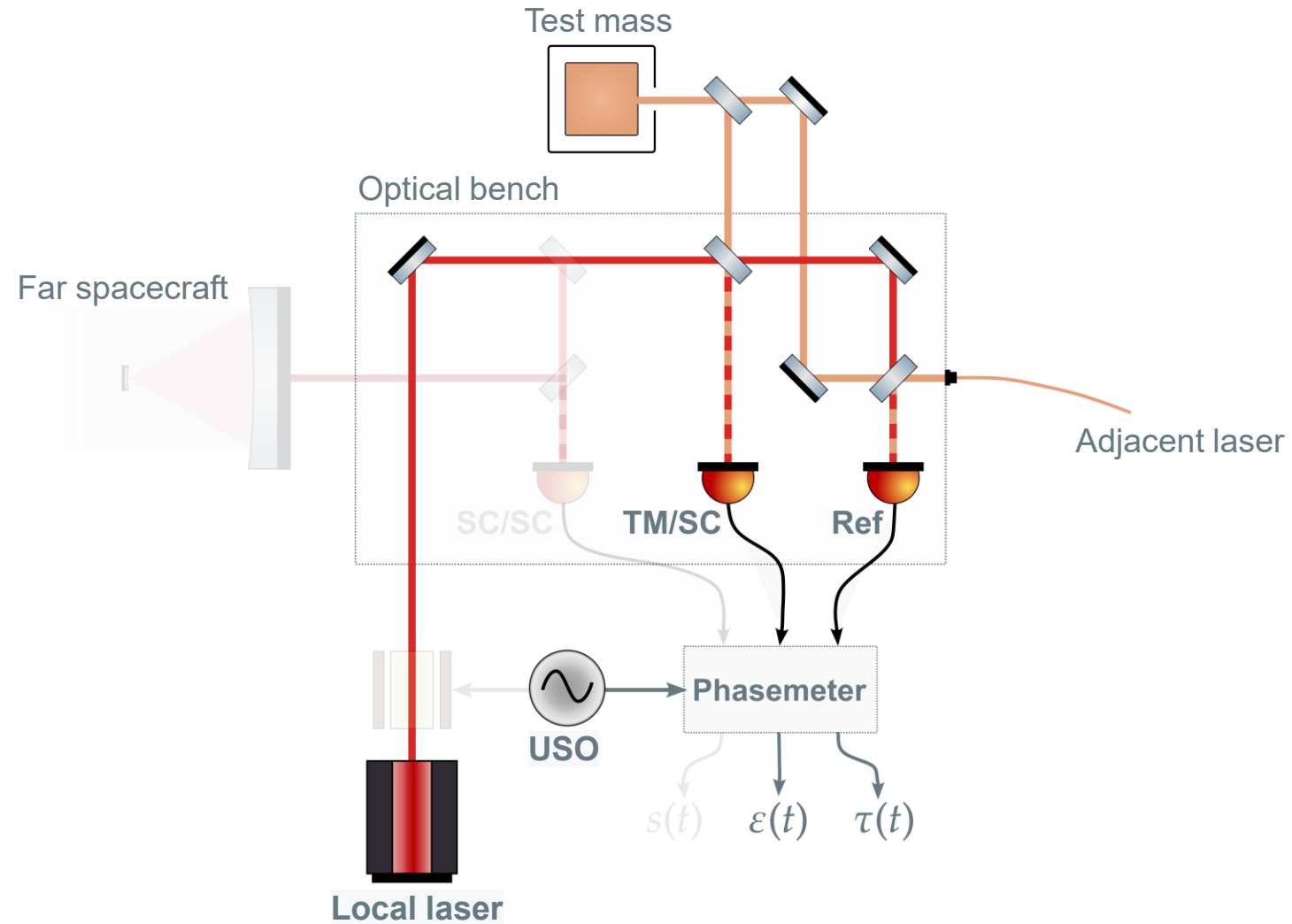
$\varepsilon(t)$ – test mass beat note

$\tau(t)$ – reference beat note

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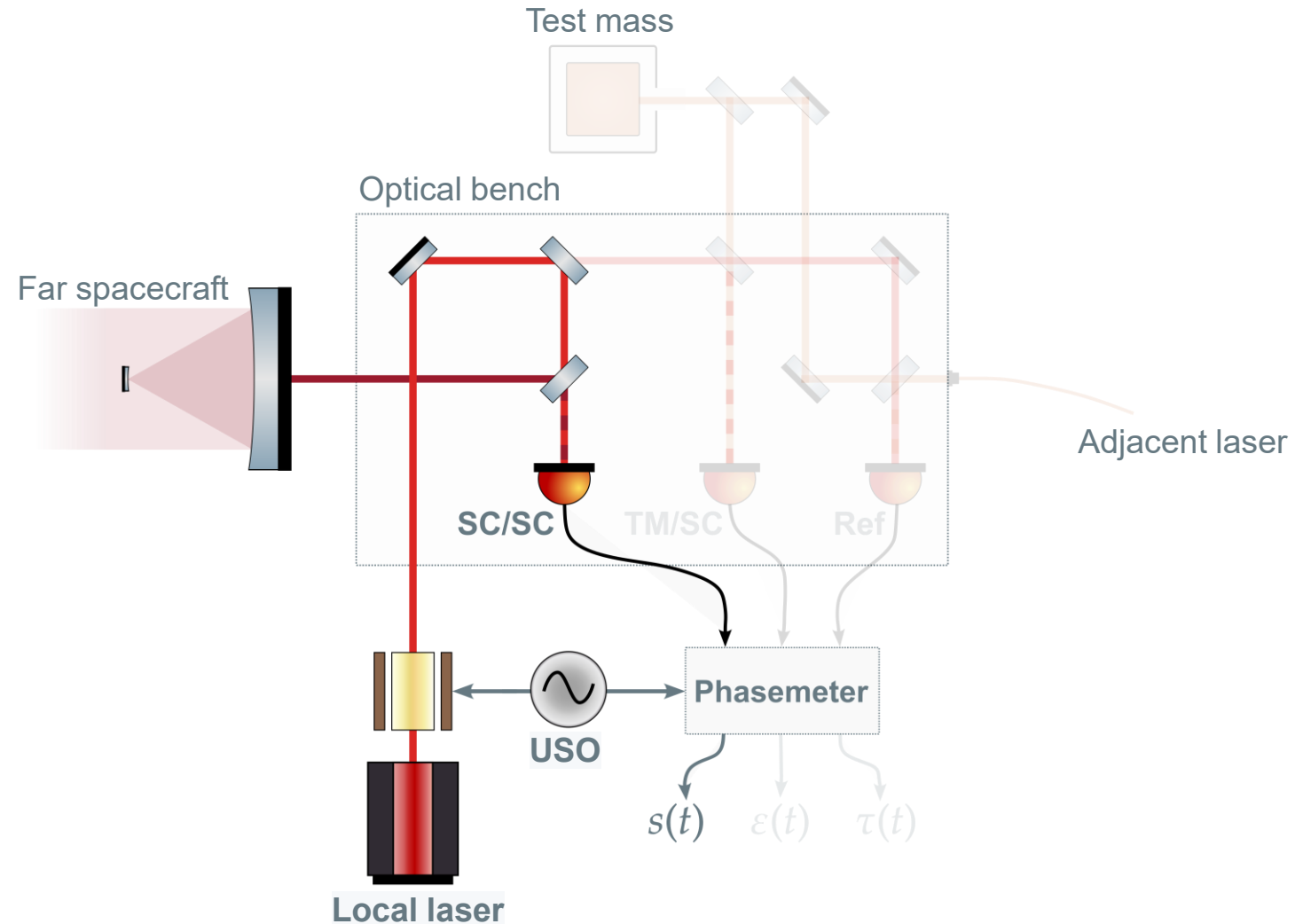
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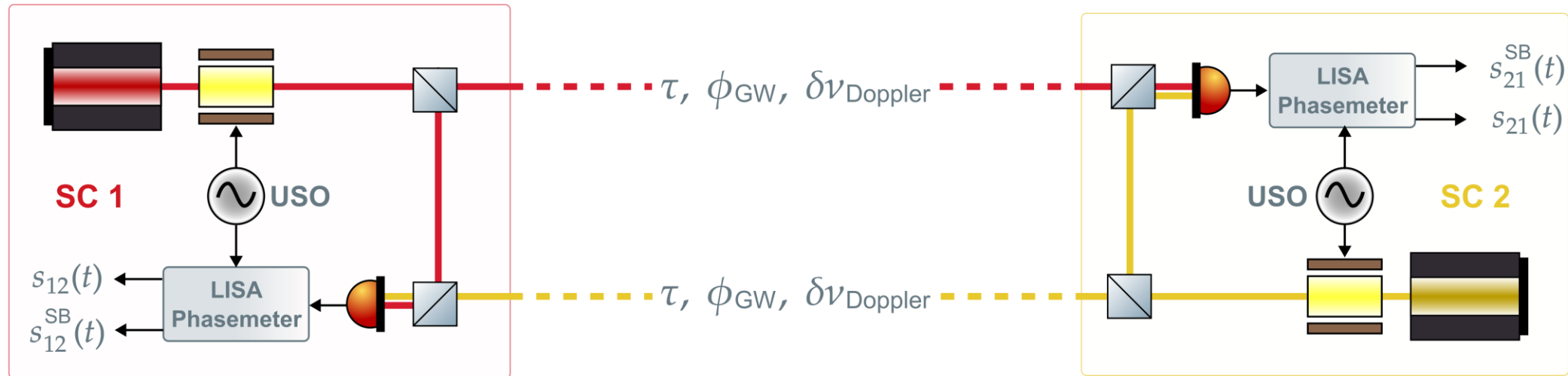
$\varepsilon(t)$ – test mass beat note

$\tau(t)$ – reference beat note

One-arm Phase Measurement



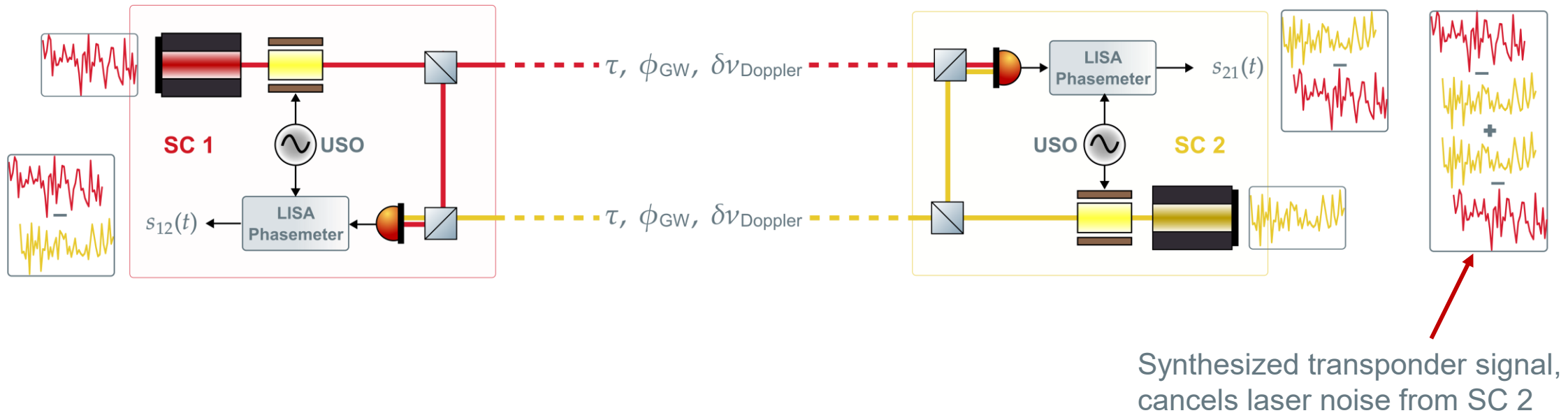
Bare-bones LISA arm



Laser Noise Reduction via TDI



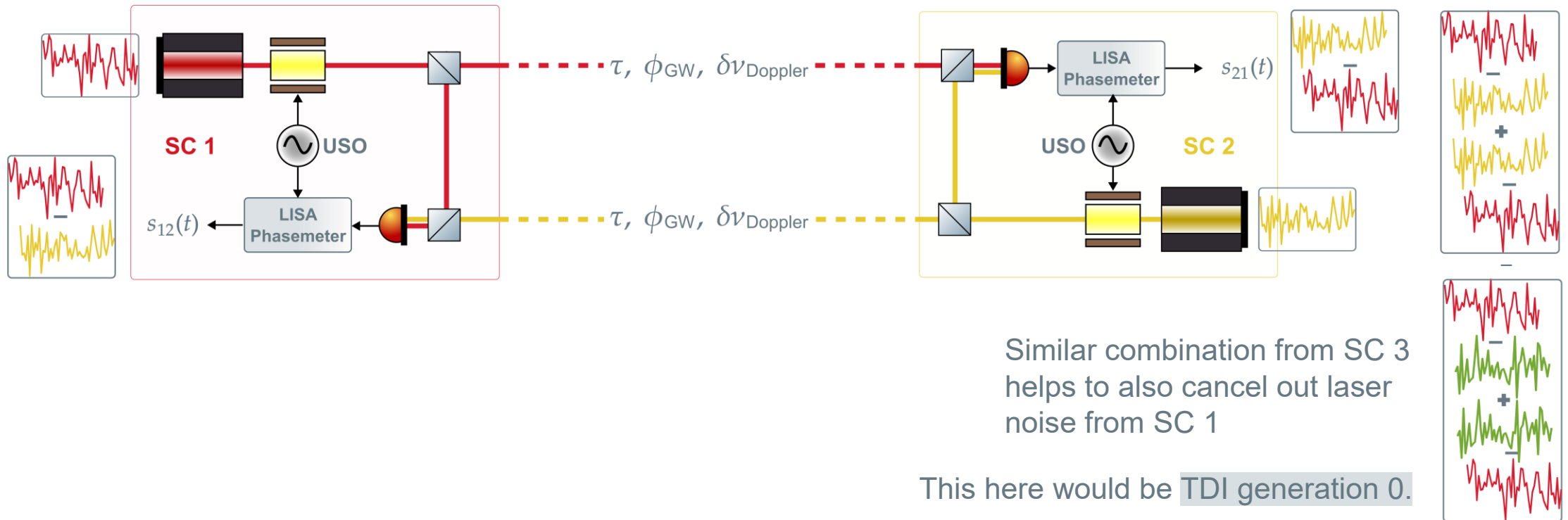
Time Delay Interferometry (TDI) is post-processing method for constructing virtual equal length arms for a LISA-like system.



Laser Noise Reduction via TDI



Time Delay Interferometry (TDI) is post-processing method for constructing virtual equal length arms for a LISA-like system.



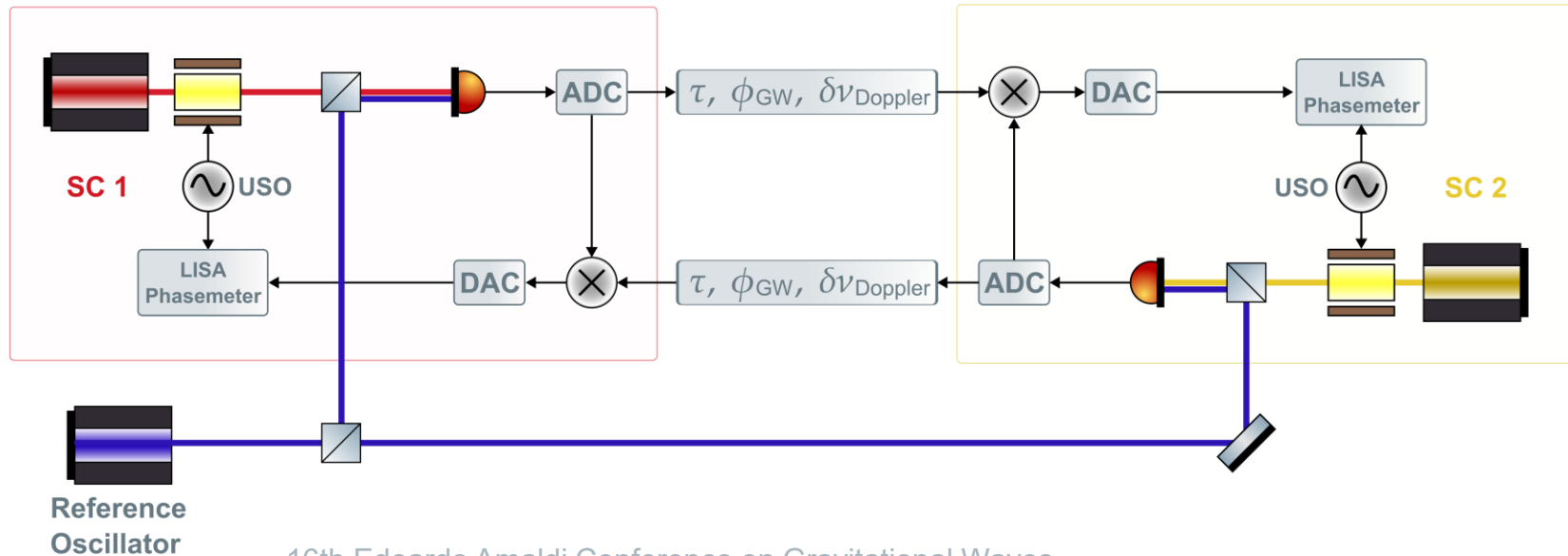
One-arm Phase Measurement

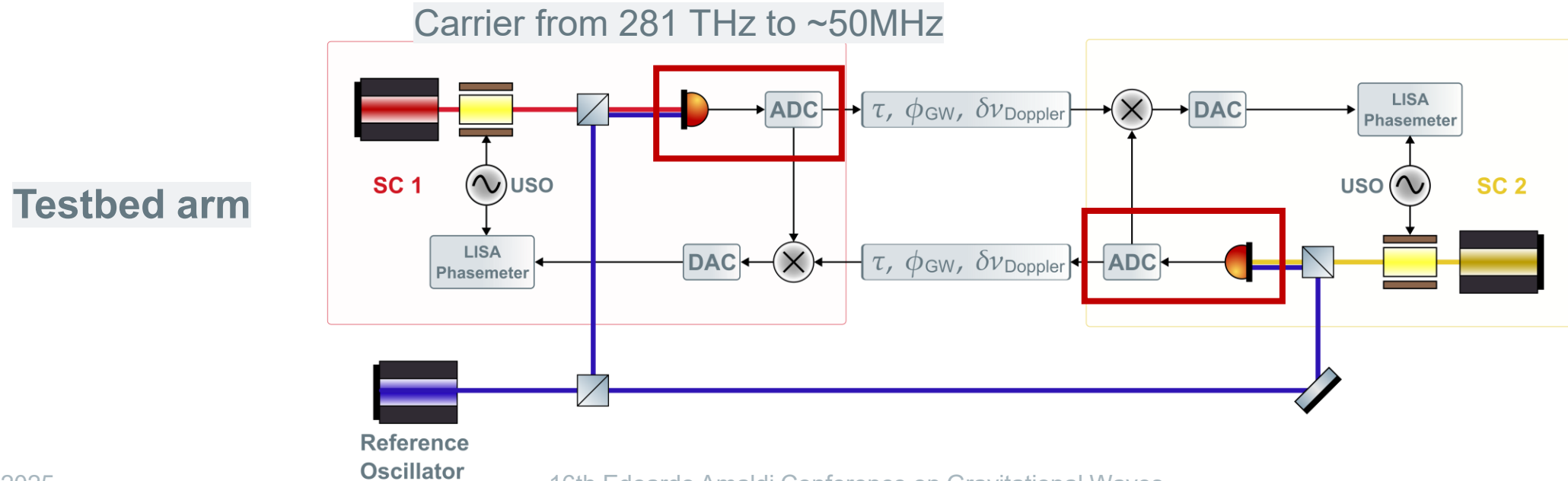
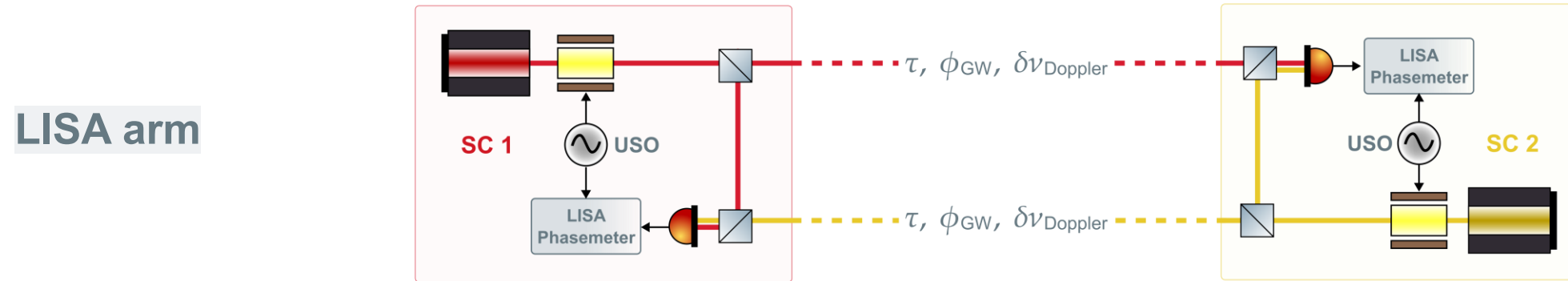


LISA arm



Testbed arm





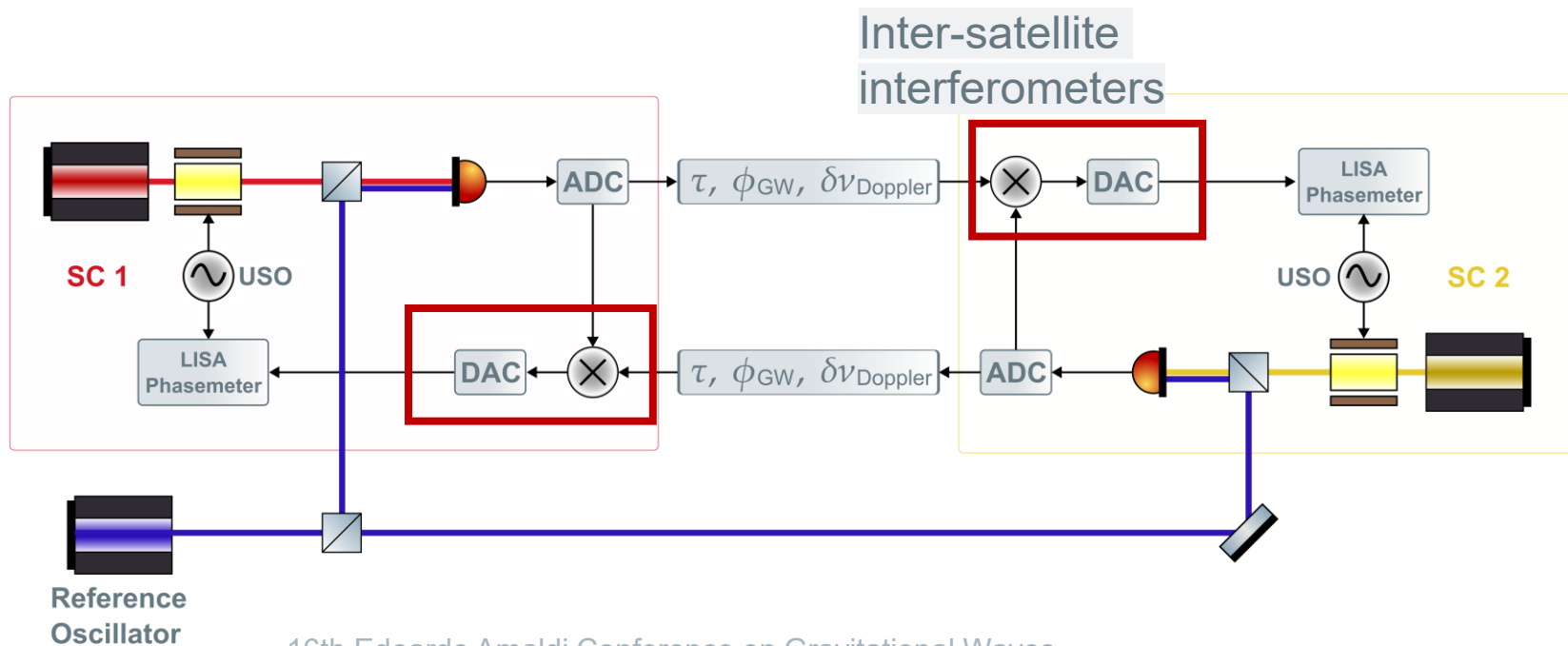
One-arm Phase Measurement



LISA arm



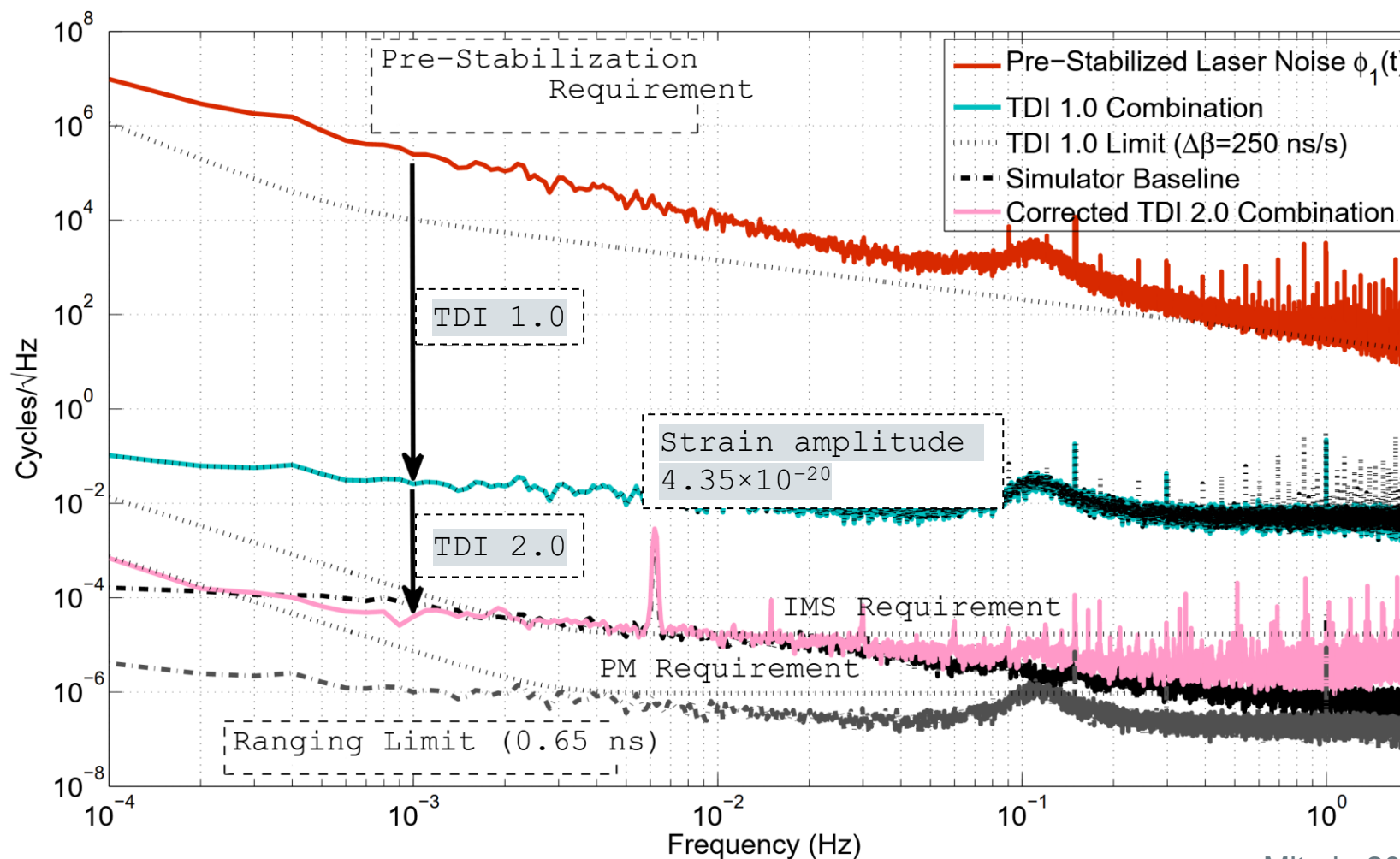
Testbed arm



Predecessor Testbed



University of Florida Laser Interferometry Simulator

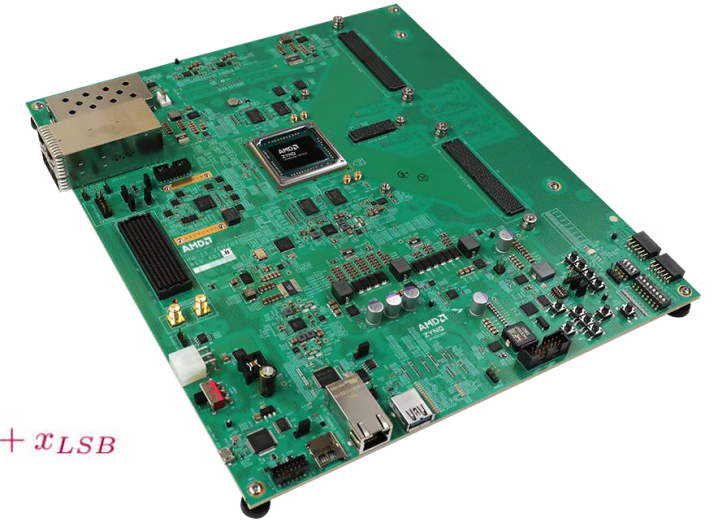
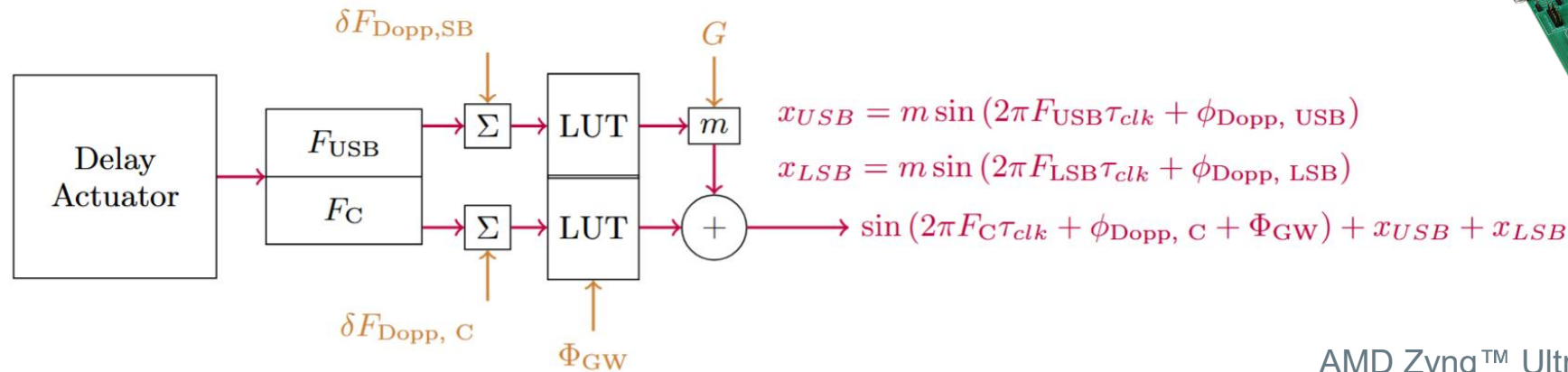


Current Status – Delay Line



A fully electronic implementation of the interferometry system together with a delay and gravitational wave injection has already been developed and is being tested.

- Time varying delays
- Doppler shifts for both the carrier and sidebands
- Gravitational wave injection



AMD Zynq™ UltraScale+™ RFSoc ZCU208 Evaluation Kit

Credit: Reid Ferguson

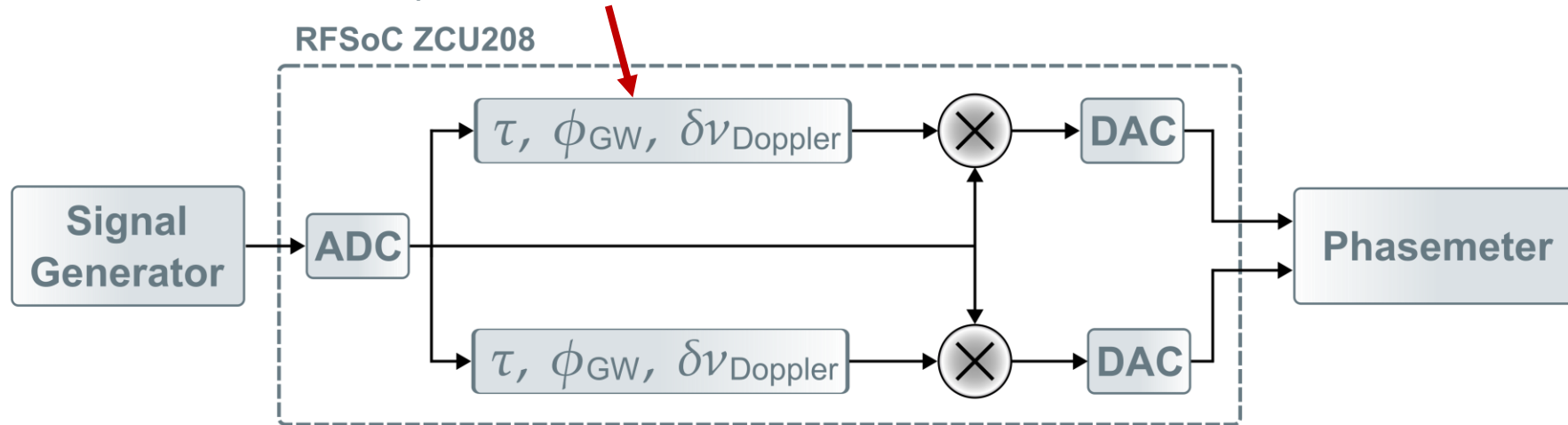
Current Status – Electrical Testbed



Latest setup for two-arm perfect transponder test

$$X_1 = (1 - \mathcal{D}_{121})\eta_{131} - (1 - \mathcal{D}_{131})\eta_{121}$$

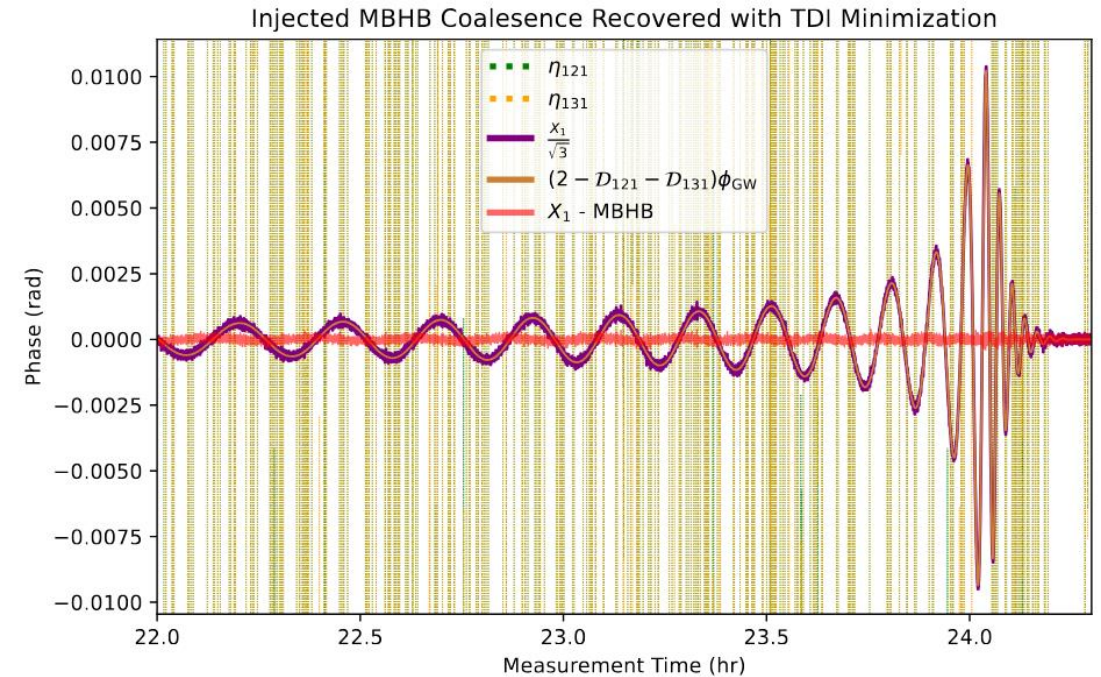
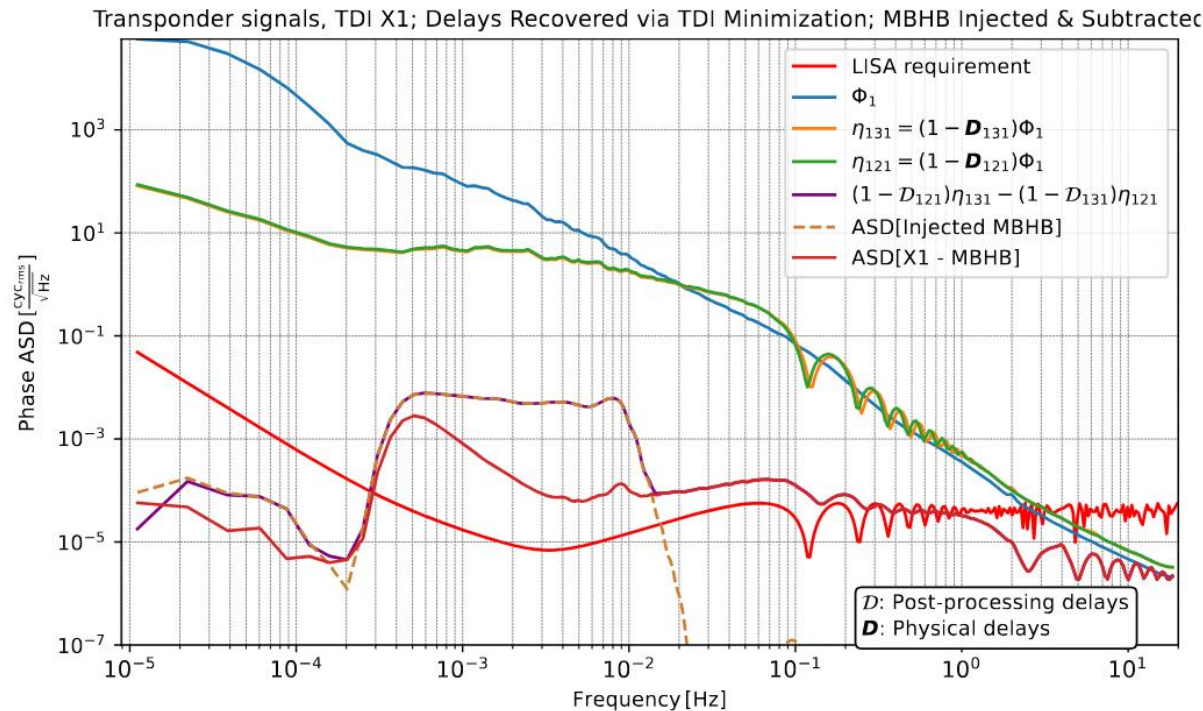
Delays, GWs and Doppler shifts are applied via a on-chip processing system which runs on a separate clock from the rest of the board



Current Status – Electrical Testbed

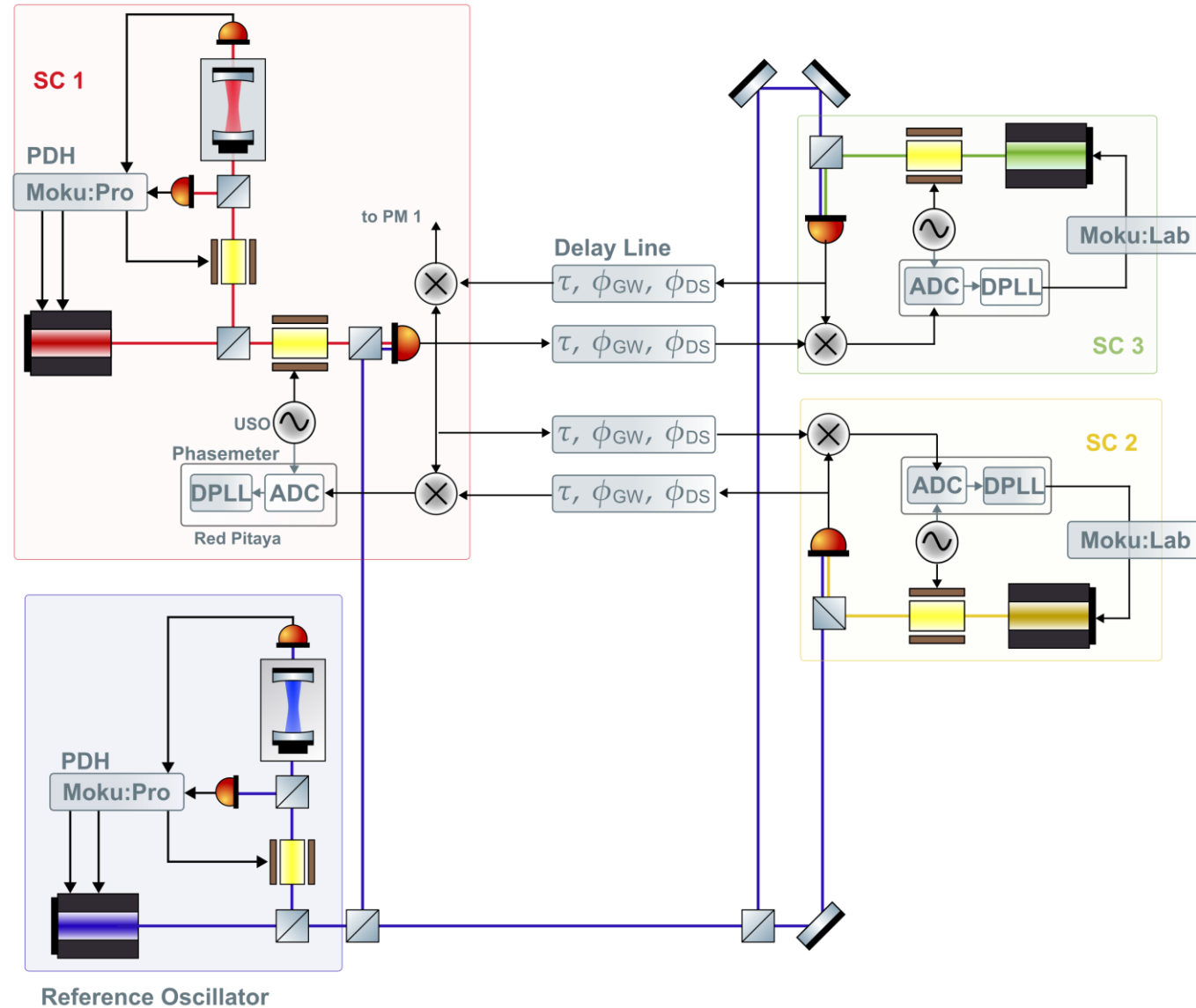


$$X_1 = (1 - \mathcal{D}_{121})\eta_{131} - (1 - \mathcal{D}_{131})\eta_{121}$$



R. Ferguson *et al.*, *Applying TDI to Hardware-Simulated Data*, 2025

Two-arm Testbed with Optical Input



Prospects and Conclusions



We have reported here on a hardware testbed to simulate the LISA signal chain together with a realistic time delay and GW injection.

- Optical input to be added

More elaborate clock system to better replicate the frequency and clock distribution system of LISA.

Addition of the third arm for better parameter estimation and data analysis

The background is a deep blue space filled with numerous small white stars. Several large, glowing galaxies are visible, including a prominent spiral galaxy in the upper left, a barred spiral in the center, and an edge-on galaxy in the lower right. A network of thin, light-colored lines crisscrosses the entire scene, representing gravitational well lines or spacetime curvature. A large, bright red triangle is superimposed on the left side, with its vertices at approximately (50, 300), (530, 550), and (110, 960). The triangle's edges are thick and have a glowing, slightly blurred appearance, with a white-to-red gradient.

Thank you for listening!